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A method for early ultrasound classification of developmental dysplasia of the hip in infants: combining YOLOv8-based keypoint detection with radiomics

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Abstract

This study introduces an innovative method for early ultrasound classification of developmental dysplasia of the hip (DDH) in infants, integrating key point detection based on the YOLOv8 model with radiomics. Leveraging YOLOv8's advanced capabilities, the method accurately identifies and locates key anatomical landmarks in pediatric hip ultrasound images. Constructing angular information through these key points serves as a crucial clinical feature for DDH classification. By fusing these clinical features with radiomic features, the method further characterizes hip joint morphology and function. The aim is to enhance the accuracy and reliability of early DDH classification, facilitating timely intervention and improving patient prognosis. Preliminary results demonstrate the feasibility and potential of this integrated approach, with an average α Angle Difference of 3.13 degrees and an average β Angle Difference of 4.83 degrees in hip DDH imaging. The model's automatic calculation of α and β angles is comparable to clinicians, alleviating clinical burden while providing valuable clinical features for DDH classification. Furthermore, based on deep learning analysis combining clinical and radiomic features, the model has achieved impressive results on multi-center test sets, with an accuracy, recall, precision, and F1 score of 0.8544, 0.8321, 0.8465, and 0.8383 respectively.

Keywords YOLOv8, Keypoint detection, Radiomics

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Introduction

Developmental dysplasia of the hip (DDH) is a common hip joint disease in infants, characterized by abnormal anatomical structures of the hip joint, leading to problems with femoral head accommodation, subluxation/dislocation, and abnormal stress. The incidence of hip instability in newborns is 1% -1.5% (5/1000 males, 13/1000 females) [1]. Early diagnosis and treatment can prevent invasive interventions and long-term complications [2]. Ultrasound is crucial for early detection of DDH and can reduce undiagnosed cases, but extensive screening brings workload and risks of human error. Deep learning can efficiently and intelligently classify/predict hip joint development by mining complex image features [3]. The feasibility of deep learning algorithms in assisting ultrasound examination for disease diagnosis has also been proven [4, 5].

However, current models exhibit certain limitations in terms of anatomical landmark localization and amalgamation of heterogeneous features.

Chen et al. [6] applied deep learning to DDH keypoint detection and achieved an angle prediction error of ±7 degrees. However, the relatively simple structure of the detection model failed to accurately extract key points. Farmmanullah Jan et al. [7] measured the acetabulum on X-rays using segmentation and keypoint detection models to assess the severity of DDH patients. Hiroki Den et al. [8] used YOLOv5 for X-ray DDH detection and classification, but also encountered bottlenecks in keypoint detection.Some scholars have conducted in-depth research on key point detection tasks [9, 10], which has improved the accuracy of detecting key points to a certain degree, but they have neglected the influence of the overall image information on classification. Maki et al. [11] merely employed radiomics analysis for the binary classification task of DDH. They successfully validated the existence of classification-related features within the medical images. However, they failed to take into account other relevant clinical information.

Liu et al. [12] have introduced a feature fusion attention network designed to capture and amalgamate both high-dimensional and low-dimensional features, offering significant insights. Building upon this foundation, we propose to harness multi-dimensional imaging texture features and integrate them with clinical attributes to collaboratively facilitate the classification task. This integrated approach holds the potential to substantially improve the accuracy and robustness of DDH subtyping, thereby advancing the precision of diagnostic outcomes. We have established several sensitive imaging indicators to assist in determining the indications for pediatric periacetabular osteotomy [13]. This study aims to establish a predictive model using deep learning networks, which detects key anatomical landmarks through ultrasound examination to assist in the early and accurate diagnosis of DDH.

Materials and methods

The research involving human participants has been reviewed and complies with the ethical standards of the institutional and national research committees. This study has obtained approval from the Ethics Committee of Children's Hospital of Soochow University (2024CS112), and informed consent was signed by all guardians of the infants.

Patients

Inclusion: (1) Patients who underwent hip ultrasound screening at Children's Hospital of Soochow University and Suzhou Municipal Hospital between January 2015 and August 2023; (2) Patients with complete imaging data and medical history records.

Exclusion: (1) Images containing a large number of annotation lines obscuring the imaging area; (2) Images with quality issues such as blurriness or incomplete ultrasound scanning areas; (3) Incomplete clinical history information.

The division of the DDH dataset is shown in Fig. 1. Retrospectively collected from January 2015 to August 2023, the experimental samples for this study consist of a multicenter dataset sourced from Children's Hospital of Soochow University and Suzhou Municipal Hospital. The experimental group comprises ultrasound images of patients with developmental dysplasia of the hip (DDH), while the control group includes images of non-DDH patients, encompassing both normal infant images and images of infants with other conditions but not DDH. Demographic data such as infant gender, age, and department of visit were collected from both hospitals. After applying the exclusion criteria, the experimental group comprised a total of 511 patients, with 465 from Children's Hospital of Soochow University and 46 from Suzhou Municipal Hospital. The control group consisted of 172 patients, including 161 from Children's Hospital of Soochow University and 11 from Suzhou Municipal Hospital. All datasets were split at the image level rather than the patient level. Each unilateral hip ultrasound image was treated as an independent sample; thus, the left and right hips of the same patient were regarded as two separate images. The Suzhou Municipal Hospital dataset was entirely reserved as an independent external test set. A formal a priori power analysis was not performed because severe DDH (Graf III/IV) is extremely rare in routine infant screening dataset. To ensure fair evaluation of model performance under class imbalance, macro-averaged metrics were used to assess classification results, as they treat all categories equally regardless of sample size.

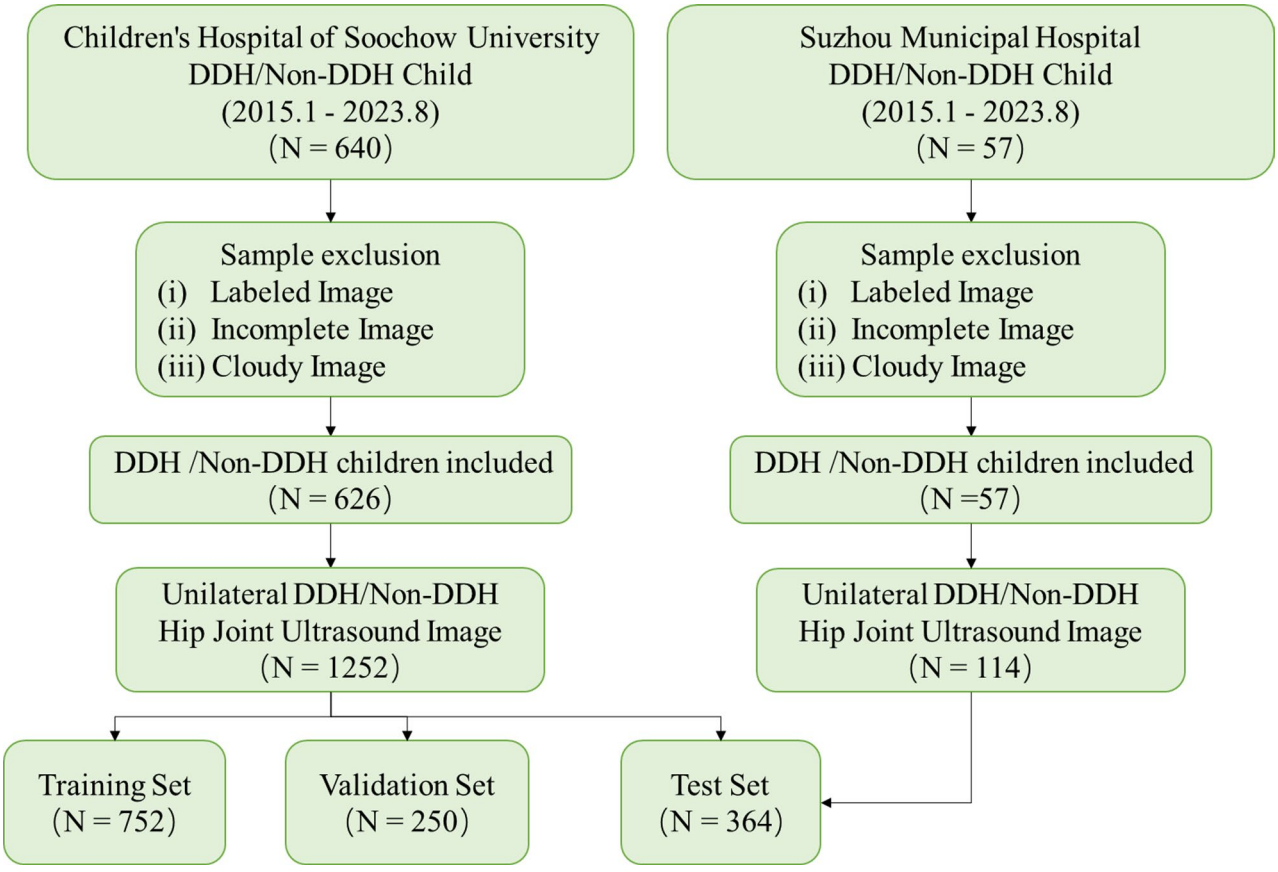


Fig. 1 Flowchart of DDH/No DDH hip joint ultrasound image used in this study. Training, validation and test sets are mutually exclusive

Image acquisition and classification standards

The ultrasound images were recorded using a 7.5 MHz linear array probe and PHILIPS ultrasound equipment (Koninklijke Philips N.V., the Netherlands). Data acquisition was performed using the Graf method, where images meeting the Graf standard plane criteria were captured from the recorded ultrasound images and used for analysis. The primary measurements taken were the α angle and the β angle. The α angle (bone roof angle) refers to the angle between the bone roof line and the baseline, while the β angle (soft tissue roof angle) refers to the angle between the soft tissue roof line and the baseline [14].

During image acquisition, patients were positioned supine with hips and knees flexed, and the ultrasound probe scanned the hip joint laterally to capture images. Ultrasound physicians with over 10 years of experience evaluated all collected images for compliance with acquisition standards and suitability for deep learning.

In our study, hip joints were classified into four types according to the Graf method, a commonly used ultrasound examination technique. Data classification was independently completed by six doctors, including physicians with 3 years of diagnostic experience and ultrasound physicians with 5 years of diagnostic experience. The classifications were then reviewed by two ultrasound physicians with over 10 years of diagnostic experience. For images with controversial classifications, the final result was adopted as the gold standard for clinical diagnosis after collective discussion. Cohen's Kappa analysis was applied to evaluate interobserver variability, and a substantial level of agreement was achieved among participating physicians, verifying the reliability of manual classification labels. Subsequently, artificial intelligence (AI) was trained to determine Graf classifications. Finally, we compared the classification accuracy between the AI system and ultrasound physicians.

According to the Graf method, a commonly used ultrasound examination technique, hip joints were classified into four types. Based on the clinical classification and diagnostic criteria, the data from Children's Hospital of Soochow University showed that the numbers of DDH I, DDH II, DDH III, and DDH IV were 663 hips, 575 hips, 9 hips, and 5 hips, respectively. Among them, DDH II was further subdivided into DDH II a, DDH II b, and DDH II c, with 218 hips, 215 hips, and 142 hips, respectively. The data from Suzhou Municipal Hospital indicated that the numbers of DDH I and DDH II were 101 hips and 13 hips, respectively. More detailed data information is displayed in Table 1.

Table 1 Demographic Information

Characteristic	Total number *n* = 683	SuZhou Municipal Hospital *n* = 57	children's Hospital of Soochow University *n* = 626
Gender	683	57(8.3%)	626(91.7%)
Male (n/%)	385	30(52.6%)	355(56.7%)
Female (n/%)	298	27(47.4%)	271(43.3%)
Age (mean ± STD, week)	6.4 W	6.3 W	6.4 W
Bilateral diagnosis	1366	114	1252
DDH Ⅰ DDH Ⅱ a	764	101	663
DDH Ⅱ b	231	13	218
DDH Ⅱ c	215	0	215
DDH Ⅲ	142	0	142
DDH Ⅳ	9	0	9
	5	0	5

Data preparation

Referring to the DDH ultrasound examination guidelines [15], we selected the following five anatomical landmarks(Fig. 2[B]): the point along the internal tangent of the iliac bone echo (landmark 1), the bony acetabular roof (landmark 2), the bony rim inflection point (landmark 3), the lower edge of the iliac branch (landmark 4), and the center point of the labrum (landmark 5). The definitions of the α and β angles are as follows: The α angle represents the included angle formed between the bony roof line (connecting landmarks 2 and 4) and the baseline (connecting landmarks 1 and 2), while the β angle denotes the included angle between the cartilaginous roof line (connecting landmarks 3 and 5) and the same baseline.

All images required cropping of the region of interest (ROI), specifically the designated hip joint area of the hip ultrasound images, to enable the training model to acquire effective information more quickly. An ultrasound physician with 10 years of diagnostic experience used LabelMe v5.3.1 (https://github.com/wkentaro/labelme) to complete the cropping of hip joint areas for both DDH and non-DDH patients. Each cropped area needed to include key anatomical landmarks. The ROI was confined strictly to the anatomical region of the infant hip joint, with irrelevant background and imaging artifacts excluded.

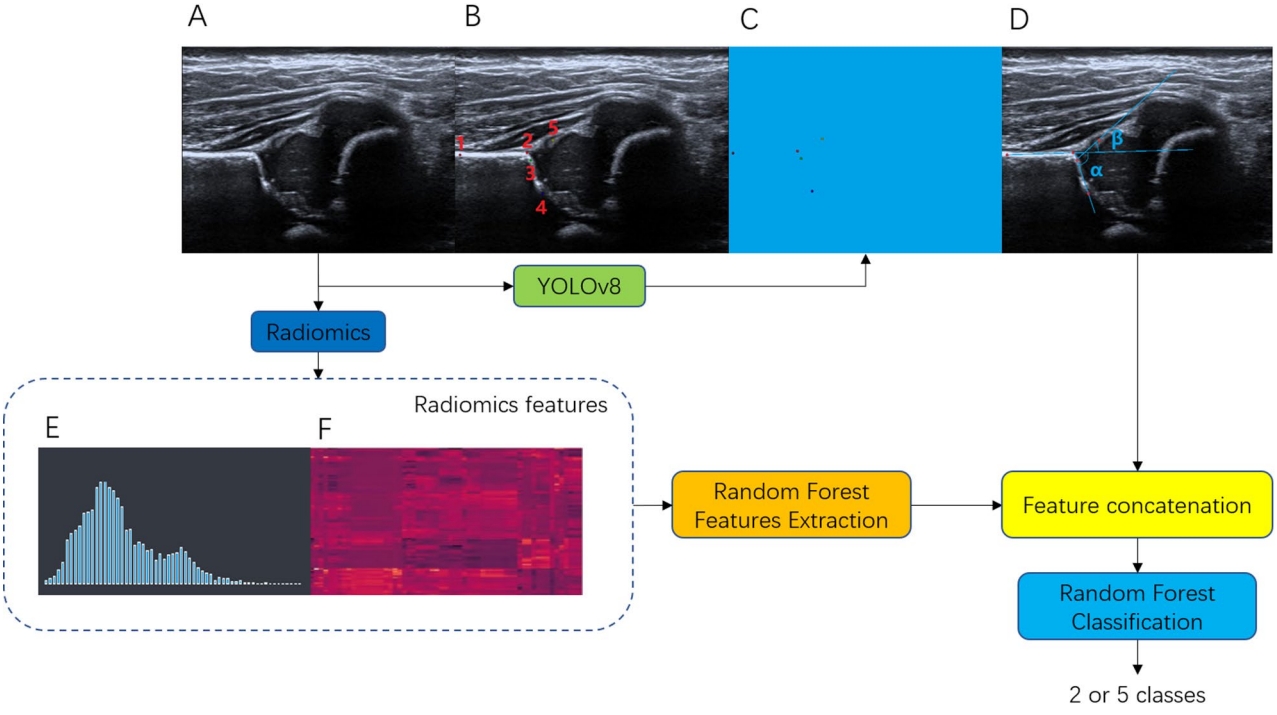


Fig. 2 Model Combining Key Point Detection Based on YOLOv8 with Radiomics. **A** Raw ultrasonography of the hip. **B** Five keypoints labeled on the hip ultrasonography based on Graf's method. **C** Five keypoints predict by YOLOv8. **D** The alpha and beta angles. **E** First-order Statistical Features. **F** Higher-order Texture Feature

The portion of the dataset from Children's Hospital of Soochow University was divided into training, validation, and test sets in a ratio of 3:1:1. Meanwhile, the data from Suzhou Municipal Hospital was entirely used as a test set, combined with the test set from Children's Hospital of Soochow University to form the final test set.

To evaluate the effectiveness of the deep learning model, three ultrasound physicians with 5 years of diagnostic experience conducted a second assessment of the test dataset, ensuring that the model's classification results could be compared with the physicians' grading results after the experiment was completed.

Methodology

The main content of the study includes: (1) Inputting the images of the experimental and control groups into a predictive model established by a deep learning network to retrospectively detect key anatomical landmarks in patients with DDH and non-DDH, comparing the detection performance of various models, and conducting a similarity comparison analysis of the clinical features α and β angles formed by these key anatomical landmarks; (2) Extracting imaging features from images of different grades through radiomics, and inputting them together with the key clinical features α and β angles into a predictive model established by a machine learning network to retrospectively predict and classify DDH patients into grades I to IV. With radiological diagnosis results as the gold standard, the performance of the model was compared based on the classification results.

This experiment utilized the YOLOv8 detection model framework for detecting key landmarks in clinical images of DDH and non-DDH. By calculating the hip α and β angles as part of the features, along with radiomic features, they participated in the multi-class classification task of DDH disease grading to seek the best-performing model. The model network architecture diagram is shown in Fig. 2.

The raw hip ultrasound image shown in Fig. 2A is processed via two independent analytical branches. The first branch implements anatomical keypoint detection, as illustrated in Fig. 2C, and subsequently computes clinical α and β angles, as depicted in Fig. 2D. The second branch performs radiomic feature extraction, including first-order statistical features presented in Fig. 2E and high-order texture features shown in Fig. 2F. Ultimately, the feature outputs obtained from the two branches are concatenated and integrated into the Random Forest model to accomplish binary classification for DDH screening and five-class grading of DDH subtypes.

YOLOv8 model

The YOLO (You Only Look Once) series of algorithms is a single-stage object detection algorithm that boasts fast detection speed and high accuracy [16]. This study utilizes the YOLOv8 detection network framework to train a method for detecting target key points in the DDH ultrasound hip region. Firstly, the parameters of a pre-trained large model are used as the initial network parameters. Then, the images from the training set are input into the pre-trained network, and the key point distance error is used as the loss function for constraint. This process aims to learn the representation of key points in the images of various grades from both the control group and the experimental group in the training set, and to provide key clinical angular features for subsequent classification tasks. (based on the publicly available code from [https://github.com/ultralytics/ultralytics/])

YOLOv11 model

Released at the end of 2024, YOLOv11 [17] is an upgraded iteration of YOLOv8 with enhanced network modules and stronger feature extraction capability. In this study, we adopted YOLOv11 as a comparative baseline for anatomical landmark detection in DDH ultrasound images. Compared with YOLOv8, it presents superior detection potential but consumes more computational resources. Meanwhile, it achieves a competitive trade-off between localization accuracy and inference speed when compared with other mainstream keypoint detection algorithms. (based on the publicly available code from [https://github.com/ultralytics/ultralytics/])

Radiomics Features Extraction(RFE)

Utilizing the functionality provided by the medical image intelligent auxiliary analysis system [18], we extracted radiomic texture features from DDH ultrasound images through radiomics. The extracted features encompassed first-order statistics ($n = 19$), shape-based 2D features ($n = 10$), gray level co-occurrence matrix (GLCM, $n = 24$), gray level run length matrix (GLRLM, $n = 16$), gray level size zone matrix (GLSZM, $n = 16$), neighbouring gray tone difference matrix (NGTDM, $n = 5$), and gray level dependence matrix (GLDM, $n = 14$). Radiomics features are normalized using z-score normalization (mean = 0, standard deviation = 1), and then feature selection is performed using random forest algorithm [19] for feature screening and modeling in ultrasound image classification prediction models. Through the feature selection process, we identified and retained the top twenty texture features with the highest correlation to DDH classification. The selected radiomic features and classification results were then used as the input and output, respectively, for training the DDH classification prediction model.

Feature fusion

The clinical angles (The alpha and beta angles) calculated from the marker points identified by the YOLO model

are considered as continuous numerical variables and normalized to 0–90 degrees. Finally, we concatenate the two features to ensure scale consistency between different modalities.

Classification module

The key clinical angular features were fused with the radiomic features, which were extracted and selected through radiomics, by concatenating them together. This fused feature set was then input into a Random Forest module for model training, using the actual clinical diagnostic classification as the gold standard for both binary classification (non-DDH vs. DDH) and five-class classification (DDH I, DDH IIa, DDH IIb, DDH IIc, DDH III+). The adjustable parameters within the Random Forest module include the number of estimators, which determines the quantity of decision trees in the forest.

Performance metrics

Table 2 below outlines the calculation equations for various metrics used to evaluate the performance of landmark detection and multi-class classification models. These include Average Angle Difference (AAD) for landmark detection, as well as Accuracy, macro Recall, macro Precision, and macro F1 for multi-class classification. The true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN) required for these calculations are derived from the confusion matrix.

Results

Diagnostic accuracy in grading of DDH

Table 3 summarizes the binary classification performance (non-DDH, DDH) of various models on the test set. The results indicate that the YOLOv8-S + RFE model achieved superior classification performance. On the training set, the data yielded an Accuracy of 0.854, a macro recall of 0.8321, a macro precision of 0.8465, and a macro F1 score of 0.8383. The classification model performance built solely using features extracted by Radiomics Feature Extraction (RFE) is inferior to that of some junior doctors, indicating that RFE may not fully capture the complexity of the data without being combined with other features or models. When clinical angle features extracted solely using the YOLO model are compared to those extracted by the YOLO model combined with RFE, the latter demonstrates superior classification accuracy that surpasses some junior doctors. This suggests that the YOLO model, and its combination with RFE, possess strong capabilities in feature extraction and classification. The features of RFE and YOLO complement each other, jointly enhancing the accuracy of the model. This combination not only exceeds the classification accuracy of some junior doctors but also surpasses the average level of junior doctors. This further proves the effectiveness of feature fusion and model combination in improving classification performance.

Table 4 presents the performance of various models on a test set for a five-class classification task (DDH Ⅰ,

Table 2 Calculation Equations for Performance Metrics

Metric	Description	Calculation Equation	Applicable Task
Average Angle Difference(AAD)	The average of the angular differences between the predicted angles and the true angles for all samples.	$\sum_{k=1}^{n} \lvert A_{Predict} - A_{True} \rvert / total$	key point detection
Accuracy	The ratio of correctly classified samples to the total number of samples.	$\sum_{k=1}^{n} a_{kk} / total$	classification
Macro Recall	The average of recall rates for each class.	$\sum_{k=1}^{n} Recall_k$ $Recall = TP / (TP + FN)$	classification
Macro Precision	The average of precision rates for each class.	$\sum_{k=1}^{n} Precision_k$ $Precision = TP / (TP + FP)$	classification
Macro F1	The F1 score calculated using the macro-average precision and macro-average recall.	$\frac{2 * macro_Recall * macro_Precision}{macro_Recall + macro_Precision}$	classification
AUC (Area Under ROC Curve)	The ROC curve has the False Positive Rate (FPR) as the horizontal axis and the True Positive Rate (TPR) as the vertical axis. AUC stands for the Area Under the ROC Curve.	$FPR = FP / (FP + TN)$ $TPR = TP / (TP + FN)$	classification

Table 3 Binary Classification Performance of Different Models or doctors on the Test Set

Models	Accuracy	Macro Recall	Macro Precision	Macro F1
RFE	0.7473	0.7094	0.7257	0.7153
YOLOv8-S	0.8379	0.8141	0.8277	0.8199
YOLOv8-S + RFE	0.8544	0.8321	0.8465	0.8383
doctor 1	0.8159	0.7491	0.8647	0.8028
doctor 2	0.8709	0.8534	0.8627	0.8580

Table 4 Five-Class Classification Performance of Different Models or Doctors on the Test Set

Models	Accuracy	Macro Recall	Macro Precision	Macro F1
RFE	0.7060	0.4033	0.4822	0.4223
YOLOv8-S	0.7115	0.4397	0.4472	0.4429
YOLOv8-S + RFE	0.7747	0.5323	0.5450	0.5296
doctor 1	0.7720	0.4663	0.6179	0.5315
doctor 2	0.8076	0.5870	0.5685	0.5776

DDH Ⅱa, DDH Ⅱb, DDH Ⅱc, DDHⅢ+). The results indicate that the YOLOv8-S + RFE model achieved superior classification performance. On the training set of the data, the accuracy, macro recall, macro precision, and macro F1 were 0.7747, 0.5323, 0.5450, and 0.5296, respectively. When constructing classification models using only features extracted by Radiomics Feature Extraction (RFE) or classifying solely with the YOLO model, their accuracy is inferior to that of some junior doctors. This indicates that, without being combined with other methods or features, these two approaches may not fully capture the complexity or clinical characteristics of the data. When the YOLO model is used in combination with the RFE method, its classification performance is comparable to that of some junior doctors. This suggests that the combination of YOLO and RFE can, to a certain extent, enhance the classification ability of the model, bringing it to a similar level as some junior doctors. At the same time, this also reflects the relatively strong subjective factors and measurement differences in the five-class classification task in actual clinical diagnosis, as there are certain variations in classification accuracy even among junior doctors.

The confusion matrix shown in Fig. 3 reveals significant clinical overlap among DDH subtypes, especially between mild cases (IIa/IIb) and normal cases (I). Due

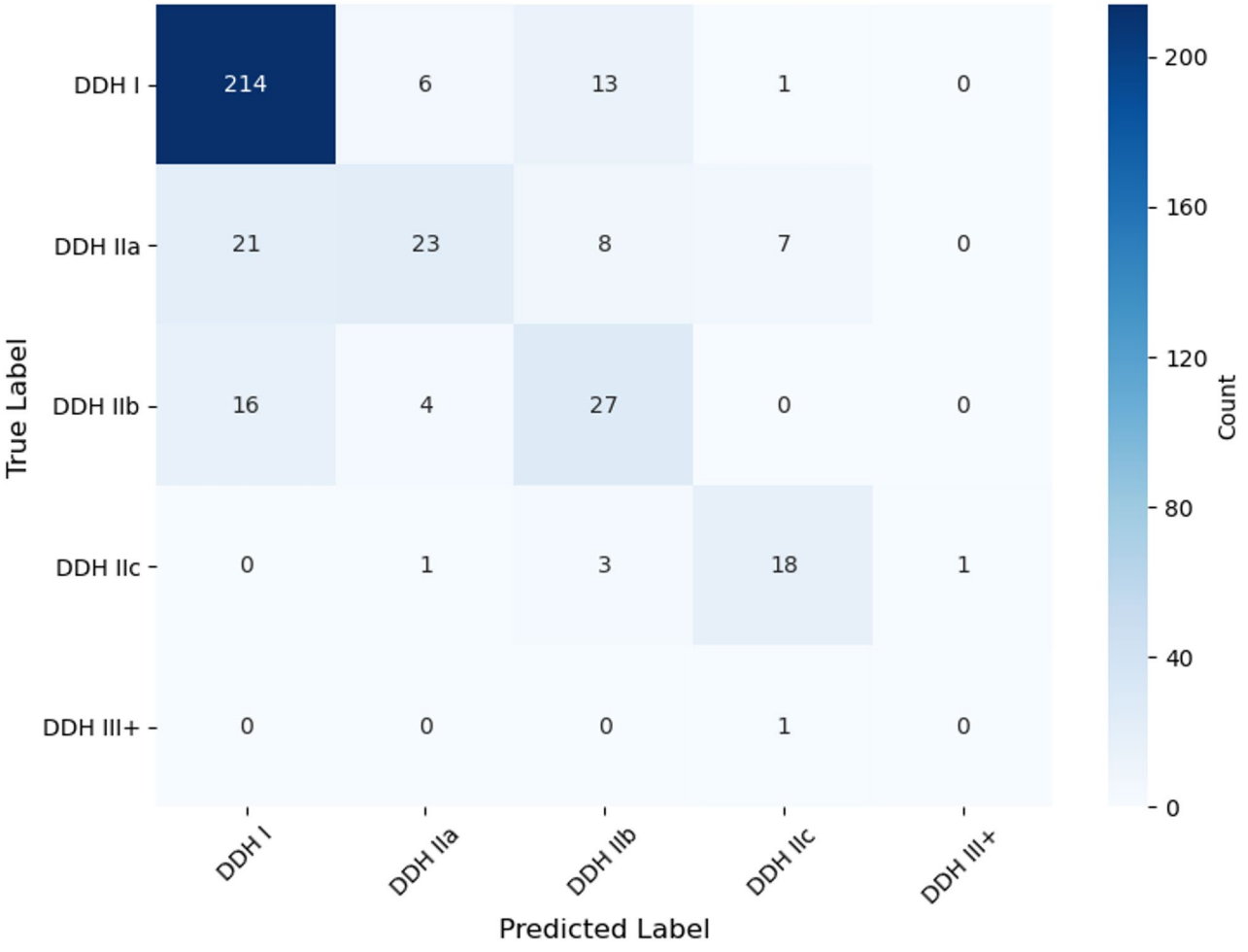


Fig. 3 Confusion Matrix for Five-Class Classification(YOLOv8-S + RFE)

to subtle α-angle variations and overlapping acetabular morphologies, 21 cases of IIa and 16 cases of IIb were misclassified as I. Furthermore, 12 cases of IIa/IIb were mutually misclassified, reflecting the ambiguity near diagnostic thresholds (e.g., α = 50°–59°). Class imbalance further distorted the outcomes, as severe DDH (III) suffered from low recall due to limited training data. This underscores the necessity for imbalance-aware techniques to enhance the detection of rare classes.

From Fig. 4, which displays the ROC curve and AUC value for binary classification, it can be observed that the AUC value is 0.8252. This indicates that the model demonstrates both effectiveness and stability in distinguishing between the two classes. An AUC value of 0.8252 suggests that the model has a relatively high ability to correctly classify instances, with a good balance between sensitivity and specificity. This highlights the model's robustness and reliability in handling binary classification tasks, making it a suitable choice for practical applications where accurate and stable predictions are crucial.

Diagnostic performance in DDH and non-DDH key point detection

Model evaluation on the validation dataset

Tables 5, 6 and 7 presents the performance of various detection models on the validation sets of both the

Table 5 Landmark Detection Performance on the Control Group of the Validation Set

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	3.86	4.24
YOLOv8-S	2.69	3.88
YOLOv8-M	3.52	4.47
YOLOv8-L	3.14	4.96
YOLOv8-X	3.20	4.68
YOLOv11-N	4.02	6.79
YOLOv11-S	4.21	6.18
YOLOv11-M	4.11	4.19
YOLOv11-L	3.32	4.18
YOLOv11-X	3.47	4.44

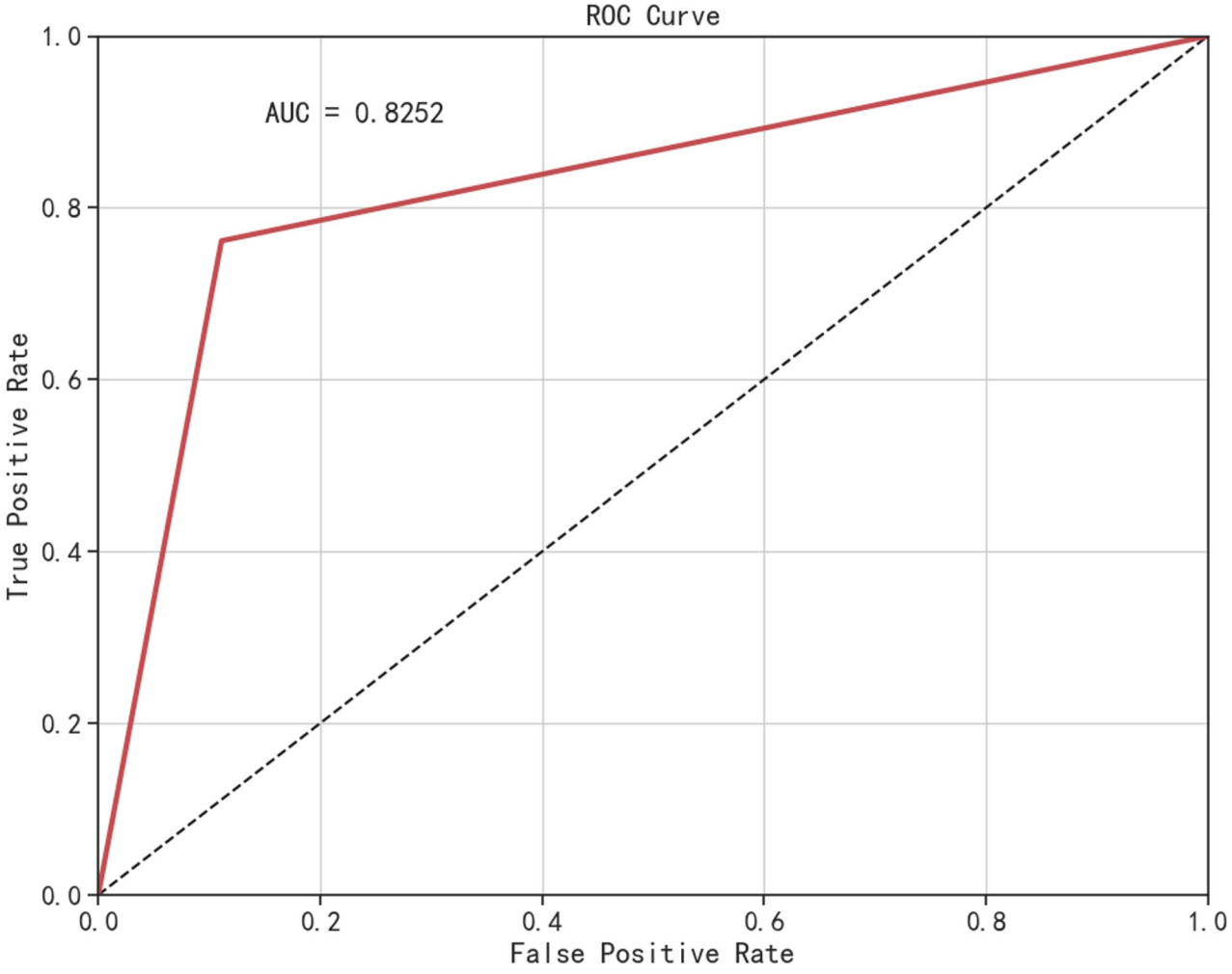


Fig. 4 ROC Curve and AUC Value for Binary Classification

Table 6 Landmark Detection Performance on the Experimental Group of the Validation Set

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	3.94	5.87
YOLOv8-S	3.64	5.90
YOLOv8-M	4.17	6.75
YOLOv8-L	3.78	6.47
YOLOv8-X	4.05	6.17
YOLOv11-N	4.13	5.98
YOLOv11-S	4.81	6.77
YOLOv11-M	3.79	6.45
YOLOv11-L	4.35	6.00
YOLOv11-X	4.01	6.56

Table 7 Landmark Detection Performance on the Validation Set

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	3.90	5.00
YOLOv8-S	3.13	4.83
YOLOv8-M	3.82	5.54
YOLOv8-L	3.44	5.66
YOLOv8-X	3.60	5.38
YOLOv11-N	4.07	6.36
YOLOv11-S	4.49	6.49
YOLOv11-M	3.96	5.39
YOLOv11-L	3.80	5.15
YOLOv11-X	3.72	5.57

Table 8 Landmark Detection Time Performance on the Validation Set

Models	time/ms
YOLOv8-N	9
YOLOv8-S	11
YOLOv8-M	27
YOLOv8-L	45
YOLOv8-X	72
YOLOv11-N	28
YOLOv11-S	32
YOLOv11-M	42
YOLOv11-L	44
YOLOv11-X	60

Table 9 Landmark Detection Performance on the Control Group of the Test Set

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	4.09	4.80
YOLOv8-S	2.74	4.17
YOLOv8-M	3.48	5.17
YOLOv8-L	3.36	5.97
YOLOv8-X	3.17	5.44
doctor 1	4.11	4.06
doctor 2	2.58	4.09

control and experimental groups. The results indicate that the YOLOv8-S model achieves superior landmark detection performance. On the overall validation set, the Average α Angle Difference is 3.13 degrees, and the Average β Angle Difference is 4.83 degrees. For the control and experimental group validation sets, the YOLOv8-S model demonstrates good landmark detection performance with Average α Angle Differences of 2.69 degrees and 3.64 degrees, respectively, and Average β Angle Differences of 3.88 degrees and 5.87 degrees. The landmark detection time increases as the complexity of the model increases.

YOLOv8-S excels in its ability to learn and capture the essential information necessary for accurate landmark detection. This is evident as it outperforms other models in terms of precision and reliability. Conversely, models with higher complexity often extract a multitude

Table 10 Landmark Detection Performance on the Experimental Group of the Test Set

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	4.49	5.39
YOLOv8-S	3.91	5.4
YOLOv8-M	4.50	6.99
YOLOv8-L	4.00	6.04
YOLOv8-X	3.93	5.90
doctor 1	5.09	6.06
doctor 2	3.87	7.01

Table 11 Landmark Detection Performance on the Test Set of the Center

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	3.88	4.92
YOLOv8-S	3.36	4.78
YOLOv8-M	4.00	5.67
YOLOv8-L	3.64	5.45
YOLOv8-X	3.61	5.12
doctor 1	4.43	5.16
doctor 2	3.00	5.43

Table 12 Landmark Detection Performance on Test Sets from Other Centers

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	5.01	5.21
YOLOv8-S	2.98	4.24
YOLOv8-M	3.51	6.15
YOLOv8-L	3.46	7.17
YOLOv8-X	3.07	6.66
doctor 1	4.54	3.95
doctor 2	3.21	4.51

Table 13 Landmark Detection Performance on Test Sets

Models	Average α Angle Difference/degree	Average β Angle Difference/degree
YOLOv8-N	4.24	5.01
YOLOv8-S	3.24	4.61
YOLOv8-M	3.85	5.82
YOLOv8-L	3.59	5.99
YOLOv8-X	3.44	5.60
doctor 1	4.47	4.77
doctor 2	3.05	5.15

of features, some of which may not be directly related to the landmark detection task, leading to a decline in overall accuracy. In terms of computational efficiency, YOLOv8-S strikes a balance between performance and speed, offering acceptable computation times that do not significantly hinder real-time applications. This makes it particularly suitable for scenarios where both accuracy and speed are crucial. Furthermore, when evaluated using data from both the experimental and control groups, YOLOv8-S consistently achieved the highest performance across all types of models. This consistency in performance underscores its robustness and adaptability, making it an ideal candidate for this specific task. Detailed computational efficiency metrics can be found in Table 8. Considering the actual clinical characteristics of DDH, the sample distribution of different subtypes is naturally imbalanced. Therefore, macro-averaged metrics were consistently adopted in this study to guarantee unbiased and fair evaluation across all categories.

Model evaluation on the test dataset

Tables 9, 10, 11, 12 and 13 summarizes the performance of various models on different test sets, all indicating that among the automatically computed models, the YOLOv8-S model exhibits superior landmark detection performance. Specifically. In detecting the α angle, the YOLOv8-S model, on the overall test set, surpasses the performance of some younger clinicians and exceeds the average level of younger clinicians. In detecting the β angle, the YOLOv8-S model demonstrates stability across multiple test sets, achieving or even exceeding the level of clinicians. The YOLOv8-S model shows a certain degree of stability in landmark detection performance across other centers. Compared to the control group, the detection difficulty in the experimental group increases, resulting in an overall increase in prediction errors for both clinicians and detection algorithms.

As shown in Fig. 5, We have carefully selected one representative case from each distinct classification of developmental dysplasia of the hip (DDH) to conduct a thorough comparison between clinicians and models,

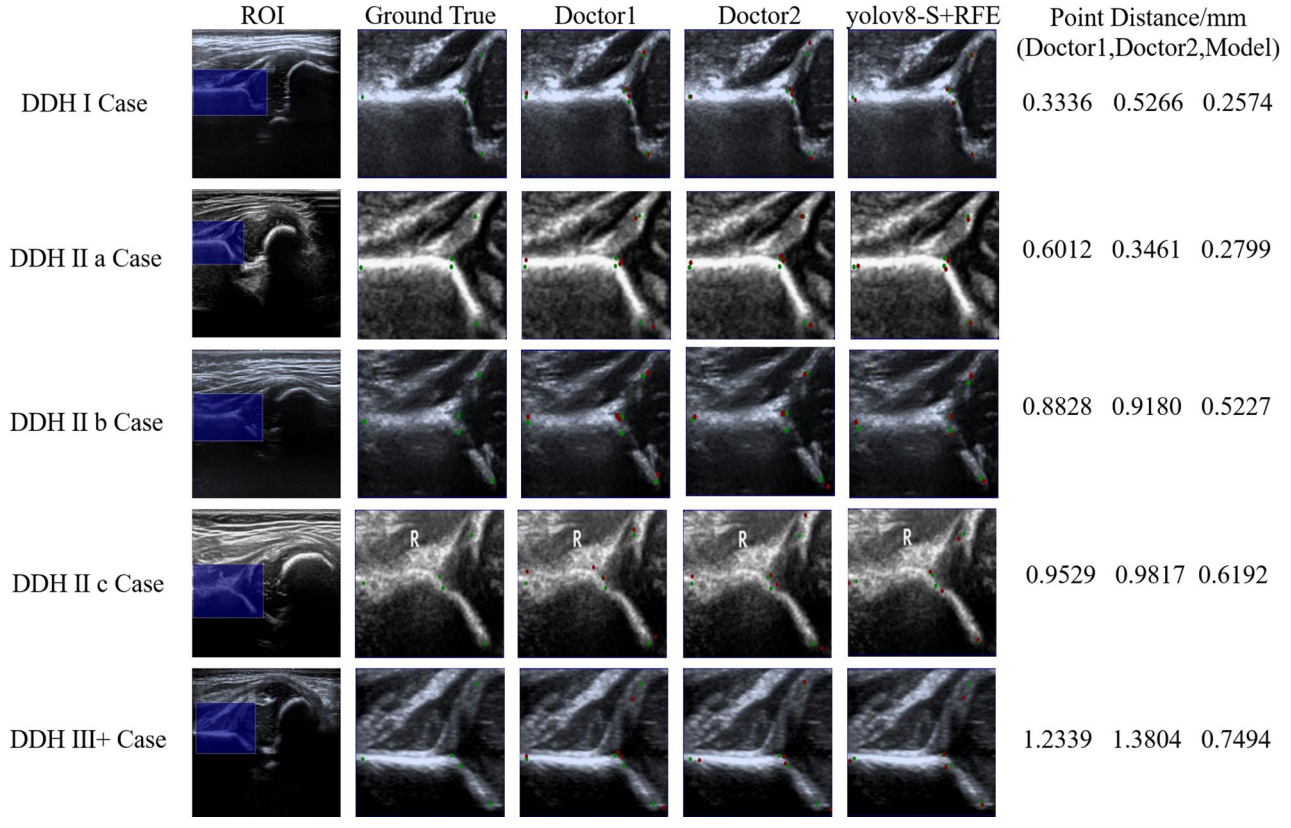


Fig. 5 Comparison of clinician performance versus model landmark accuracy across different classifications of developmental dysplasia of the hip (DDH)

focusing on the accuracy of labeled landmark positions relative to their actual locations. Our findings reveal several notable insights. Firstly, the YOLOv8-s + RFE model exhibits remarkable performance in point prediction, showing a closer approximation to Ground Truth compared to the clinicians' assessments. Secondly, as the severity of DDH escalates, the task of identifying crucial anatomical landmarks becomes increasingly challenging, which is reflected in a steady rise in the average distance between the landmarks identified by both the model and the clinicians. This highlights the complexity and variability associated with more severe cases of DDH. Lastly, we observed significant discrepancies in error rates among different clinicians, indicating that there is considerable inter-individual variation in landmark identification skills. Furthermore, the positioning of landmarks also varies across different DDH classifications, suggesting that the nature and severity of the dysplasia impact the clinicians' ability to accurately pinpoint these critical anatomical features.

Conclusion

The key point detection in hip DDH imaging reveals an Average α Angle Difference of 3.13 degrees and an Average β Angle Difference of 4.83 degrees. The automatic calculation of α and β angles, along with the average distance error and average angular error, is comparable to that of clinicians. This model not only alleviates the clinical burden but also provides clinical characteristics for DDH classification. The deep learning-based grading analysis of DDH, utilizing clinical features and radiomics of the hip joint, achieves an Accuracy, macro Recall, macro Precision, and macro F1 of 0.8544, 0.8321, 0.8465, and 0.8383 respectively on the test set. This demonstrates its applicability in clinical classification and prediction of DDH, proving that deep learning and artificial intelligence technologies can effectively assist in clinical decision-making. Experiments have shown that the combination of key point detection and holistic image feature extraction can effectively enhance the accuracy of DDH classification. In terms of interpretability, the key point detection module offers a certain degree of interpretability, while the features extracted from radiomics can be analyzed to a certain extent. In terms of computational speed, the detection model and classification model deliver results within seconds, enabling their direct use as preliminary clinical results. This can save doctors' operation time and improve diagnostic efficiency.

Discussion

The present study evaluated the performance of various models in classifying developmental dysplasia of the hip (DDH) based on clinical features and radiomics, with a focus on the integration of YOLOv8-S and Radiomics Features Extract (RFE) methods. The results demonstrate the advantages and limitations of our approach, as well as potential directions for future improvements.

Advantages

Effective feature extraction and integration

The YOLOv8-S + RFE model exhibited superior classification performance in both binary and five-class classifications of DDH. This suggests that the combination of clinical angle features extracted by YOLO and radiomic features is complementary and enhances the model's accuracy. This integration not only surpassed the classification accuracy of some junior doctors but also exceeded their average level, demonstrating the effectiveness of feature fusion and model combination in improving classification performance.

Robust and reliable binary classification

The ROC curve and AUC value of 0.8252 for binary classification indicate that the model is both effective and stable in distinguishing between the two classes. An AUC of 0.8252 represents good discriminative power for differentiating DDH from non-DDH cases, confirming that the model possesses satisfactory sensitivity and specificity for auxiliary clinical screening. The AUC value suggests that the model has a relatively high ability to correctly classify instances, achieving a good balance between sensitivity and specificity. This highlights the robustness and reliability of the model in handling binary classification tasks, making it a suitable choice for practical applications where accurate and stable predictions are crucial.

Reduction in label detection time

Although the time required for label detection increases with the complexity of the model, the time consumed is still much less than the actual labeling time in clinical practice. This may be a limitation in the actual clinical setting, where rapid and accurate diagnosis is often required. Therefore, the reduced detection time achieved by the model may represent a clinically significant and feasible application.

Furthermore, misclassification of DDH subtypes has tangible clinical implications. Missed or delayed diagnosis of moderate and severe DDH may postpone early intervention and increase the risk of irreversible hip developmental deformity. By contrast, false-positive results may trigger unnecessary parental anxiety and redundant repeated ultrasound examinations. Our YOLOv8-S + RFE model helps reduce subjective diagnostic variation, standardize DDH grading, and improve the reliability and efficiency of clinical infant hip screening.

The numerical trends presented in Tables 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 and 13 consistently confirm that the YOLOv8-S + RFE fusion model yields superior

performance in both anatomical landmark detection and DDH subtype classification. Across internal validation, external multi-center test sets, control and experimental dataset, the proposed fusion framework outperforms standalone radiomics and single YOLO models, and achieves competitive diagnostic accuracy compared with junior clinicians. These tabulated quantitative results further support the robustness and clinical practicality of our integrated method.

Limitations

Complexity in five-class classification

While the YOLOv8-S + RFE model performed well in binary classification, its performance in five-class classification was relatively lower. This may be due to the increased complexity and subjectivity in distinguishing between different DDH subtypes, even among junior doctors. The results suggest that further improvements are needed to enhance the model's ability to capture the nuances and clinical features of each subtype. Additionally, formal a priori statistical power analysis was not conducted before model construction. The primary reason is that severe Graf III/IV DDH cases are inherently rare in routine infant hip ultrasound screening, with only a small number of such patients enrolled in this real-world retrospective dataset. The limited sample size of high-grade DDH subtypes inevitably leads to class imbalance and may weaken the model's generalization and recall performance for these clinically critical rare cases. To mitigate evaluation bias caused by imbalanced sample distribution, we adopted macro-averaged metrics throughout all classification assessments to ensure equal evaluation weight for each DDH subtype.

This retrospective study mainly focused on model performance comparison and multi-center applicability verification; statistical confidence interval (CI) analysis was not included in the current experimental design. We will supplement CI-based statistical verification and larger prospective dataset validation in future research to further strengthen the statistical credibility of the model.

Future improvements

Enhanced feature extraction

To improve the model's performance in five-class classification, further research is needed to develop more sophisticated feature extraction methods that can better capture the clinical features and nuances of each DDH subtype. This may involve integrating additional data sources, such as patient demographics, medical history, and genetic information, to provide a more comprehensive view of the disease.

Validation and generalization

Future studies should validate the model's performance on larger and more diverse datasets to ensure its generalizability and robustness. Additionally, comparisons with other state-of-the-art methods and clinical evaluations by experts are needed to further assess the model's clinical utility and potential for integration into routine clinical practice. While our multi-center design and triple-radiologist validation support robustness, limited replicates constrain variability analysis. Future work will incorporate 5 + repeated runs to quantify performance stability, ensuring stronger generalizability claims.

Overcoming classification challenges

There is significant clinical ambiguity in the DDH five-class classification between subtypes II a/II b/II c, which exhibit overlapping alpha angle ranges. This difference leads to differences between observers and label noise, especially in critical situations. In addition, category imbalance exacerbates performance bias, as rare subtypes (such as Class III) are underrepresented, leading to a decrease in the recall rate of clinically critical cases that require timely intervention. To address these challenges, future research will integrate advanced techniques, including label smoothing (LS) to reduce overconfidence in fuzzy prediction, category reweighting to prioritize underrepresented categories, and stratified sampling to ensure a balanced category distribution during model training. These strategies aim to enhance robustness and clinical applicability by improving subtype differentiation and fair performance across all DDH categories.

In conclusion, the present study demonstrates the potential of integrating YOLOv8-S and RFE methods for DDH classification based on clinical features and radiomics. While the approach has shown promising results in binary classification, further improvements are needed to enhance its performance in five-class classification and reduce marker detection time. Future research should focus on developing more sophisticated feature extraction methods, optimizing the model's architecture, and validating the model's performance on larger and more diverse datasets.

Authors' contributions

Zhenhong Zhu and Yi Zheng are co-first authors.Zhenhong Zhu and Yi Zheng wrote the main manuscript text.Ting Feng and Yanhong Lu processed the data from their respective hospitals.Jisu Hu and Xusheng Qian prepared Tables 1, and 2. Zhiyong Zhou prepared Tables 9, 10, 11, 12 and 13. Yakang Dai and Xiaodong Wang provided technical guidance.All authors reviewed the manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
The research involving human participants has been reviewed and complies with the ethical standards of the institutional and national research committees. This study has obtained approval from the Ethics Committee of Children's Hospital of Soochow University (2024CS112), and informed consent was signed by all guardians of the infants.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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